Lycée LA FAYETTE Clermont-Ferrand

CPGE

Analyse harmonique des SLCI TD 1

Compétences visées: B2-08, C2-02, E3-06, E3-07 v1.2

Lycée La Fayette – 21 Bd Robert Schuman – 63000 Clermont-Ferrand – Académie de Clermont-Ferrand

Diagrammes de Bode : tracé et identification de FT

1 Exercice 1 : Diagrammes asymptotiques de Bode

Tracer les diagrammes asymptotiques de Bode des fonctions de transfert ci-dessous :

Question 1 $F_1(p) = \dfrac{10}{(1+0,5p)(1+0,25p)}$

Question 2 $F_2(p) = \dfrac{10(1+0,25p)}{(1+0,5p)p}$

Question 3 $F_3(p) = \dfrac{100\,000(2+10p)}{(10\,000+80p+p^2)}$

2 Exercice 2 : Diagrammes asymptotique et réel de Bode

Question 1 Tracer le diagramme asymptotique de Bode de la fonction de transfert :

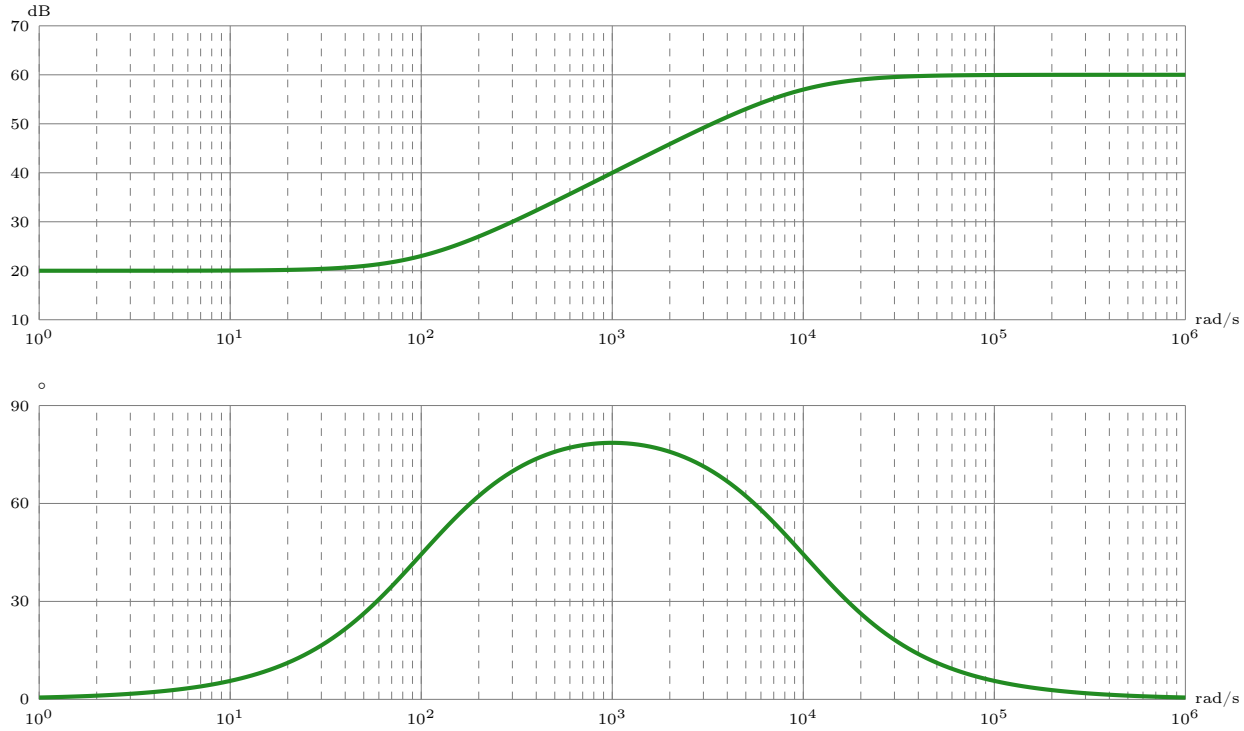
$$H(p) = \frac{58(p+1,1)}{p(4+p)(p+8)}$$

Question 2 En déduire l'allure du lieu réel de transfert de Bode.

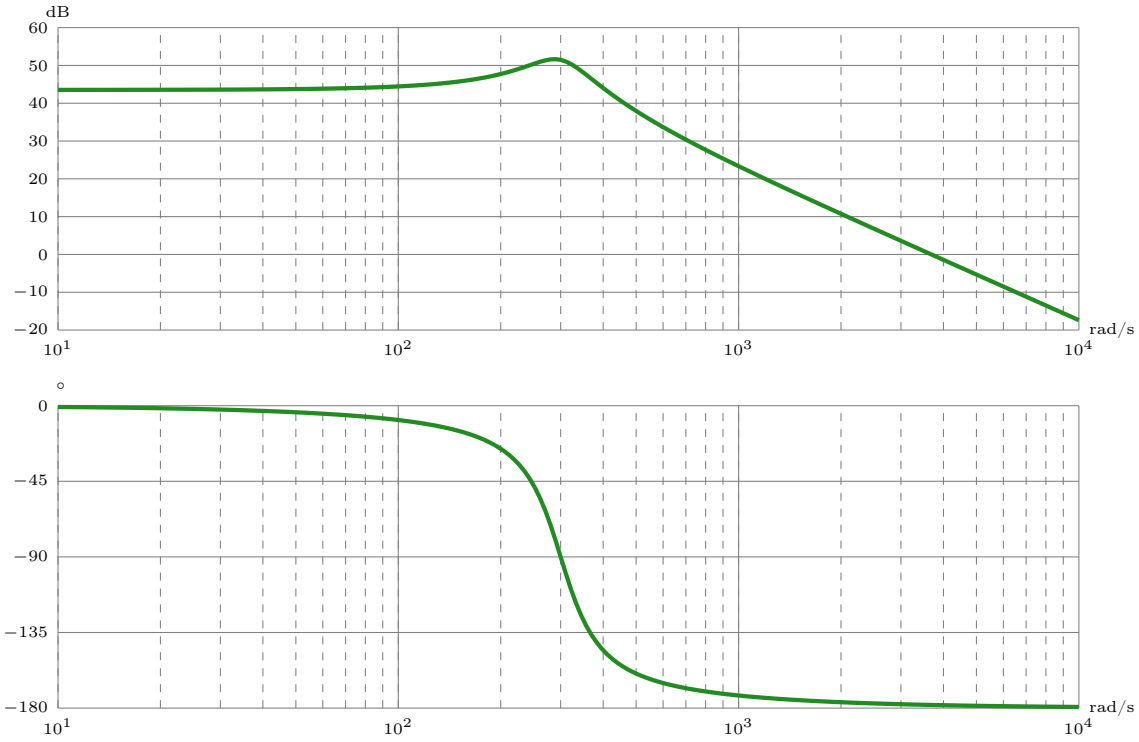
3 Exercice 3 : Identification fréquentielle - Modèle de comportement

Identifier un modèle de comportement sous la forme d'une fonction de transfert dans le domaine de Laplace à partir des réponses fréquentielles ci-dessous :

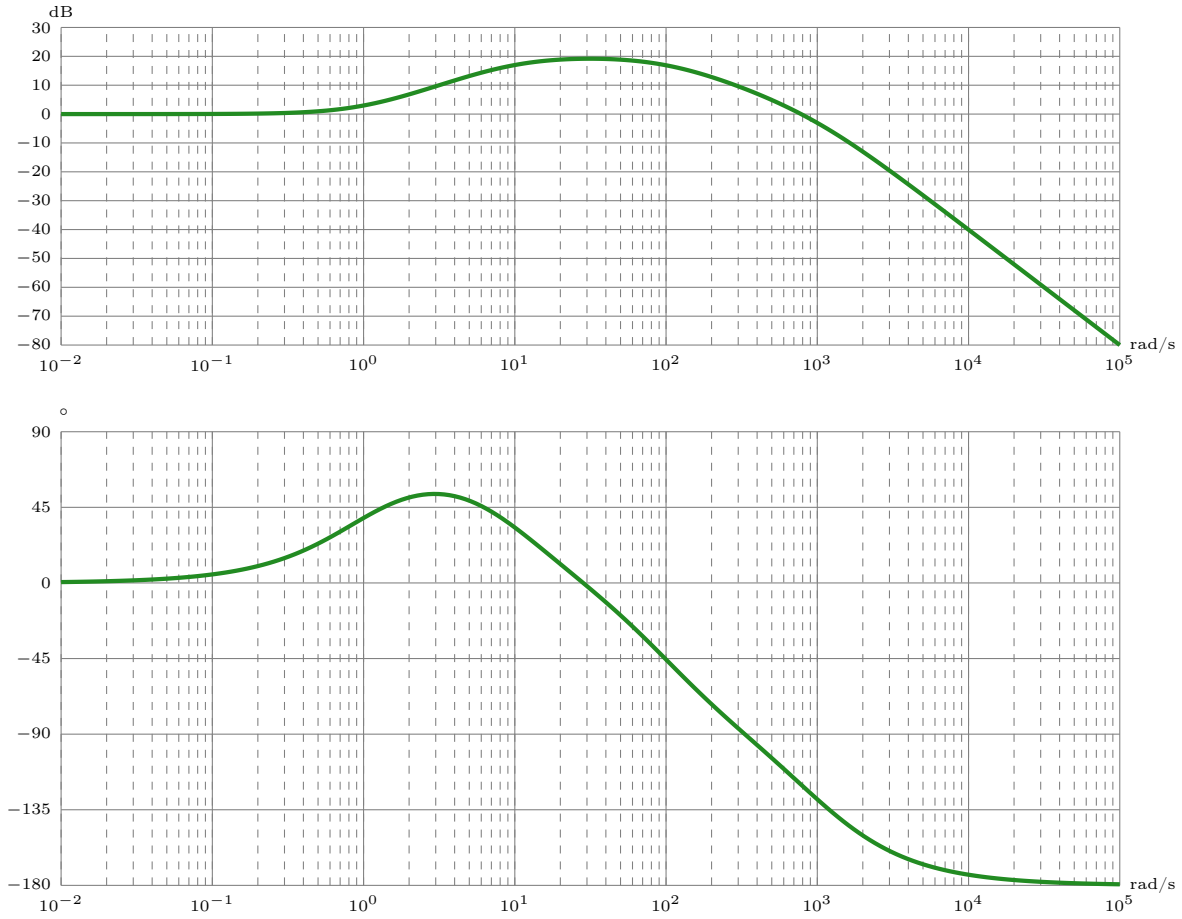
Question 1



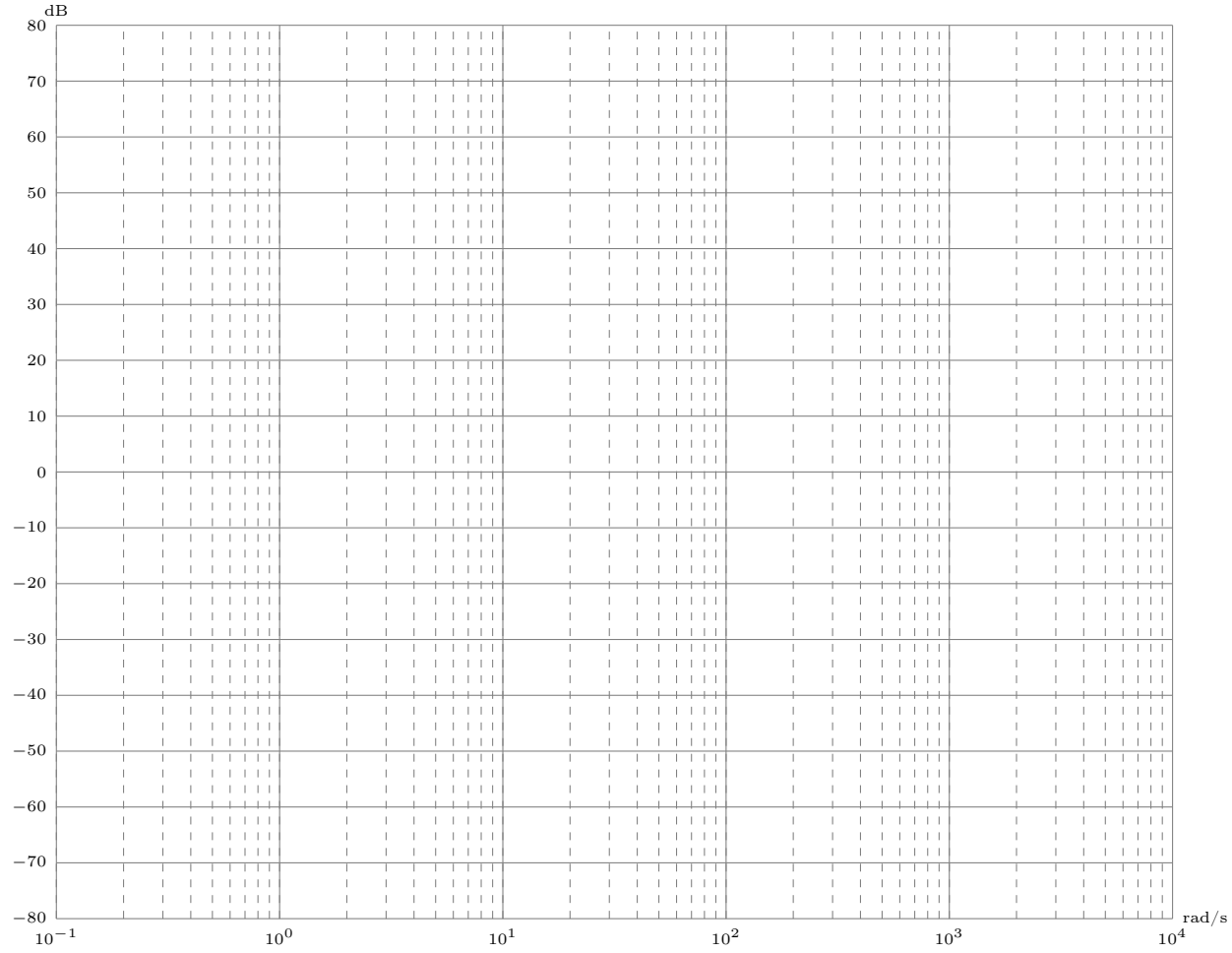
Question 2

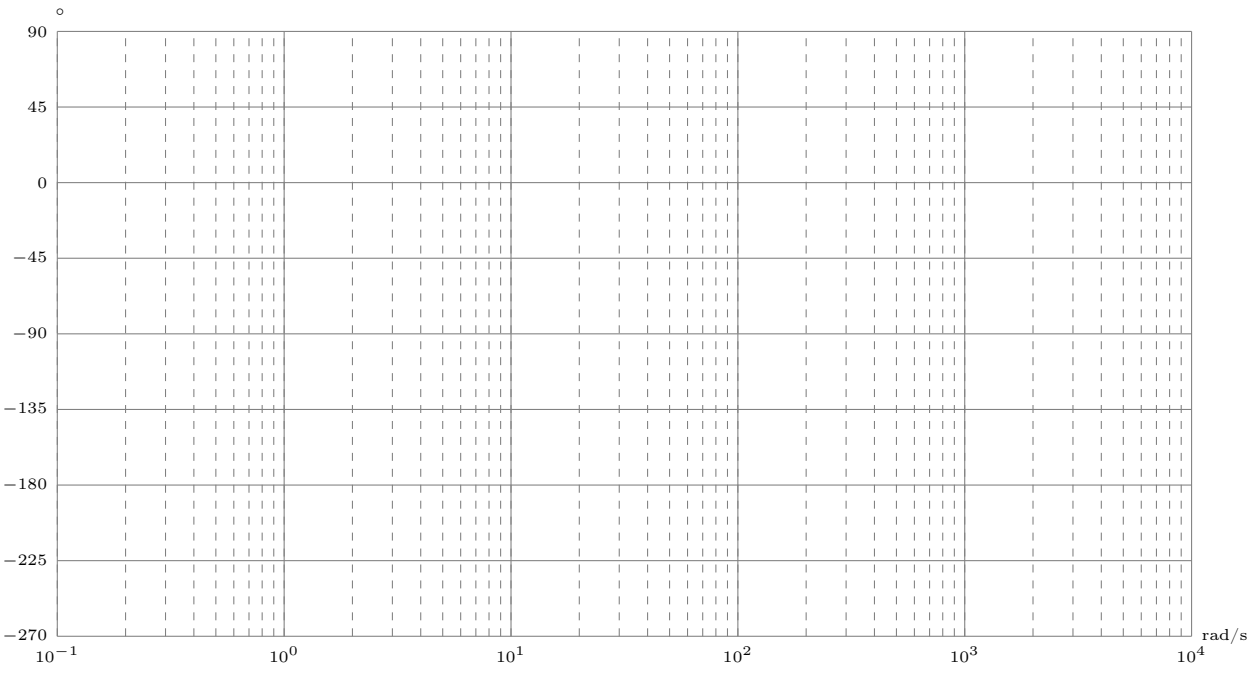


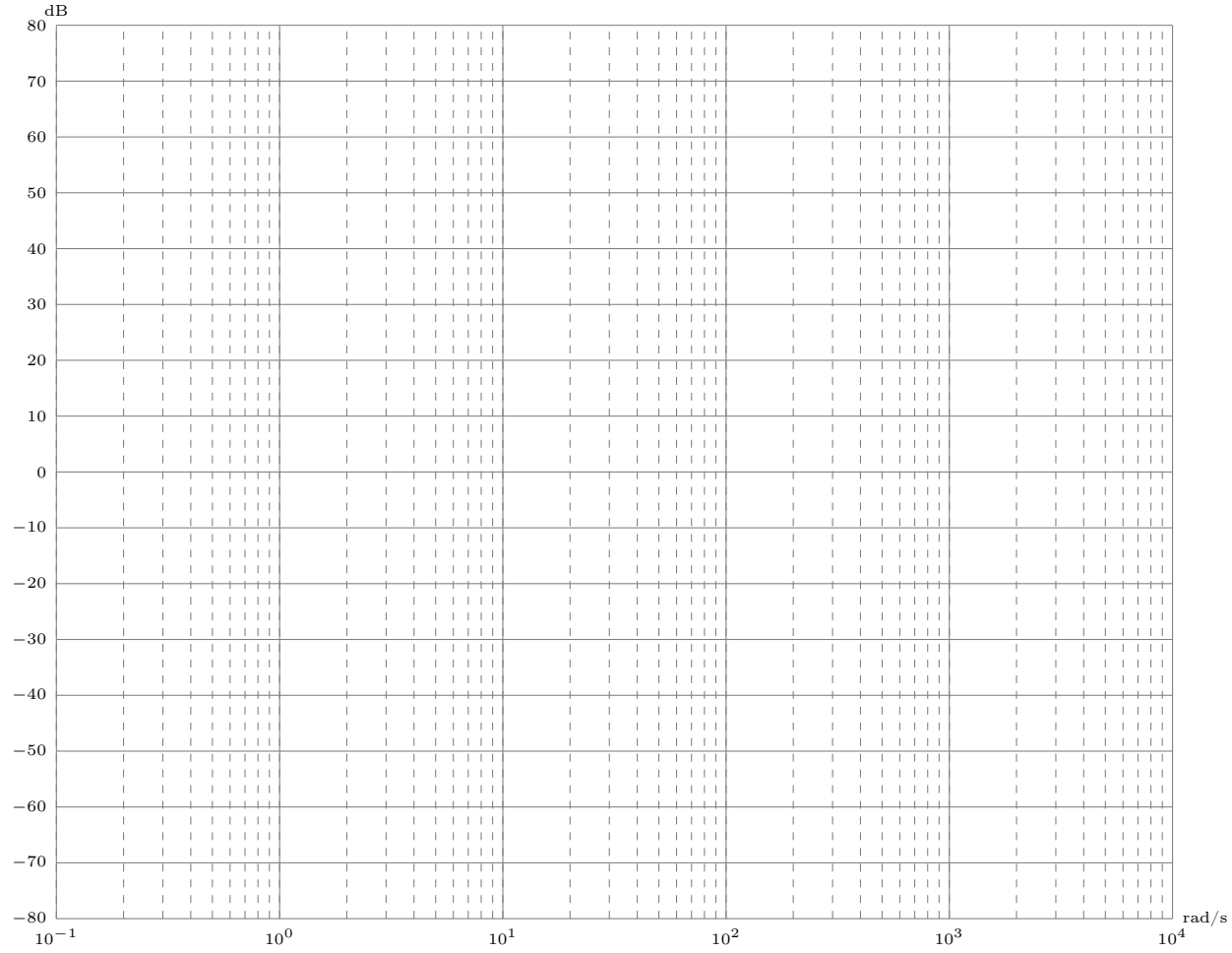
Question 3

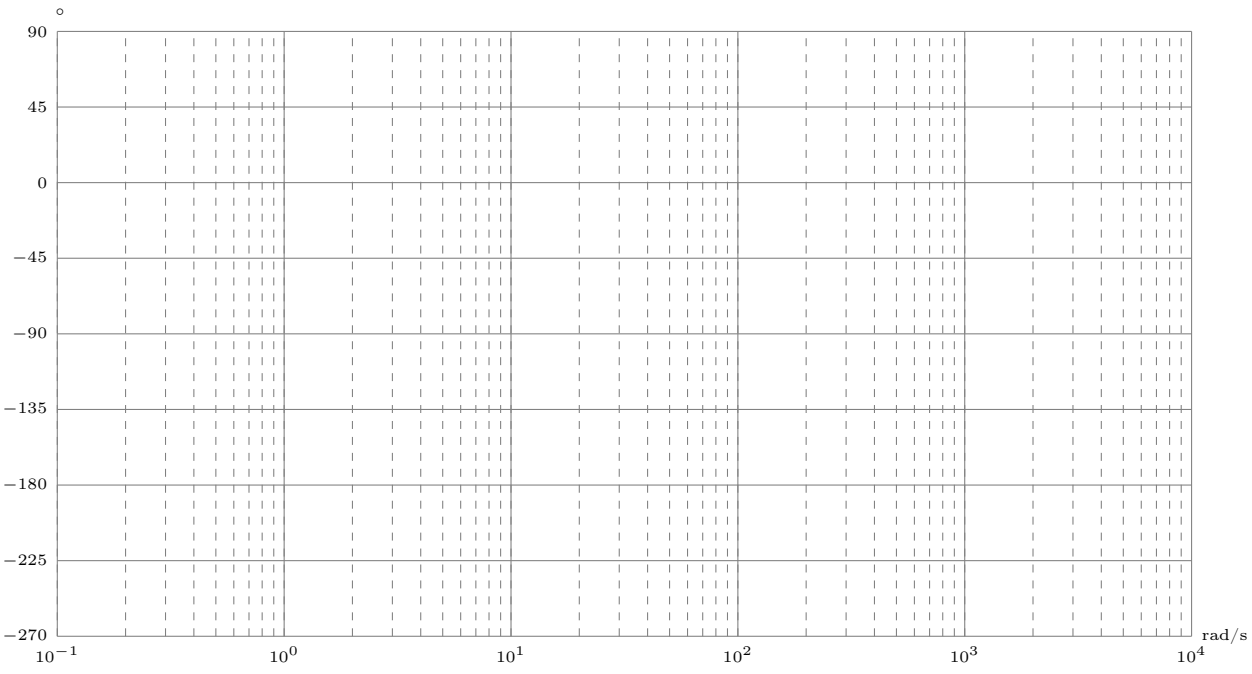


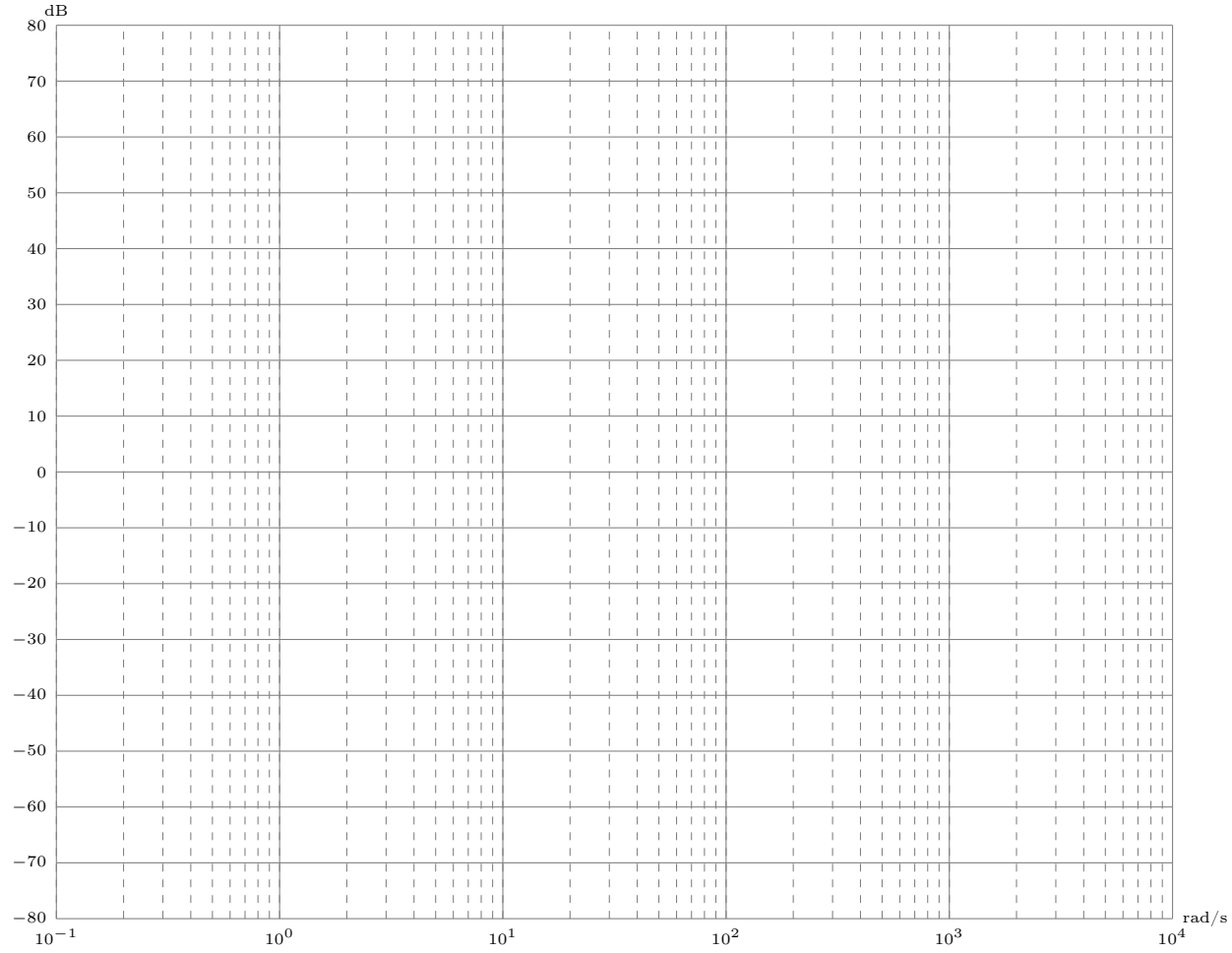
Question 4 Pour le second lieu de transfert de l'exercice 3 (question 2), déterminer la marge de gain et de phase.

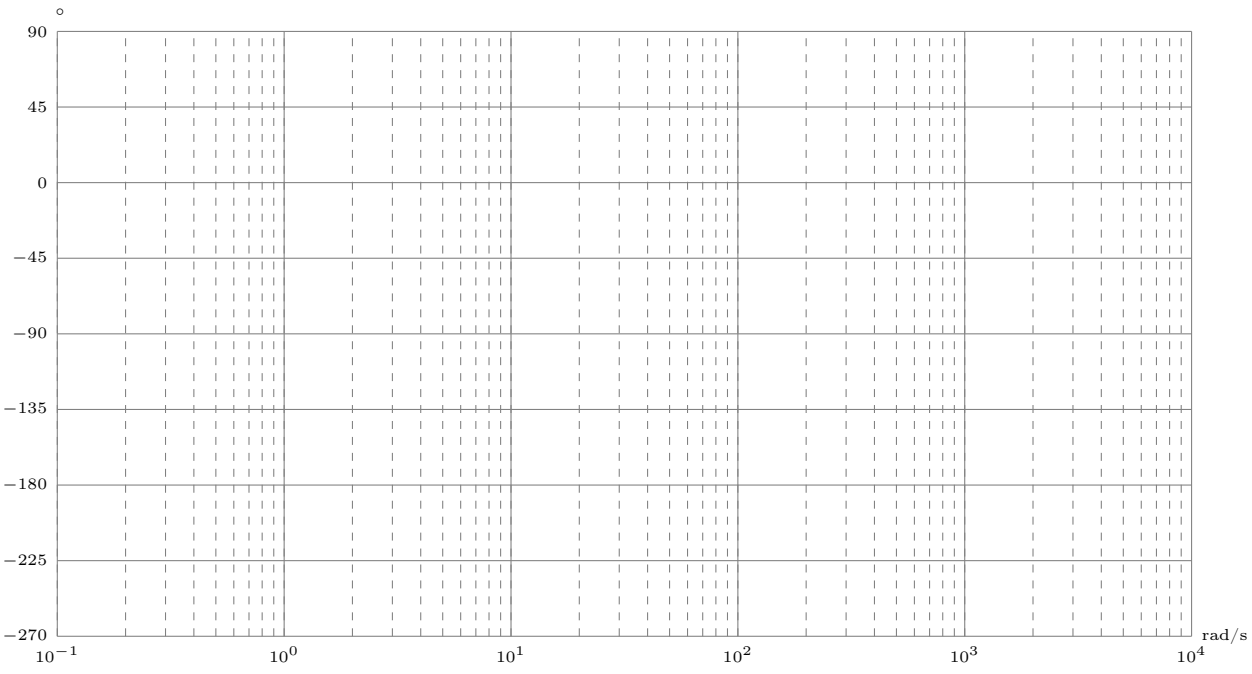

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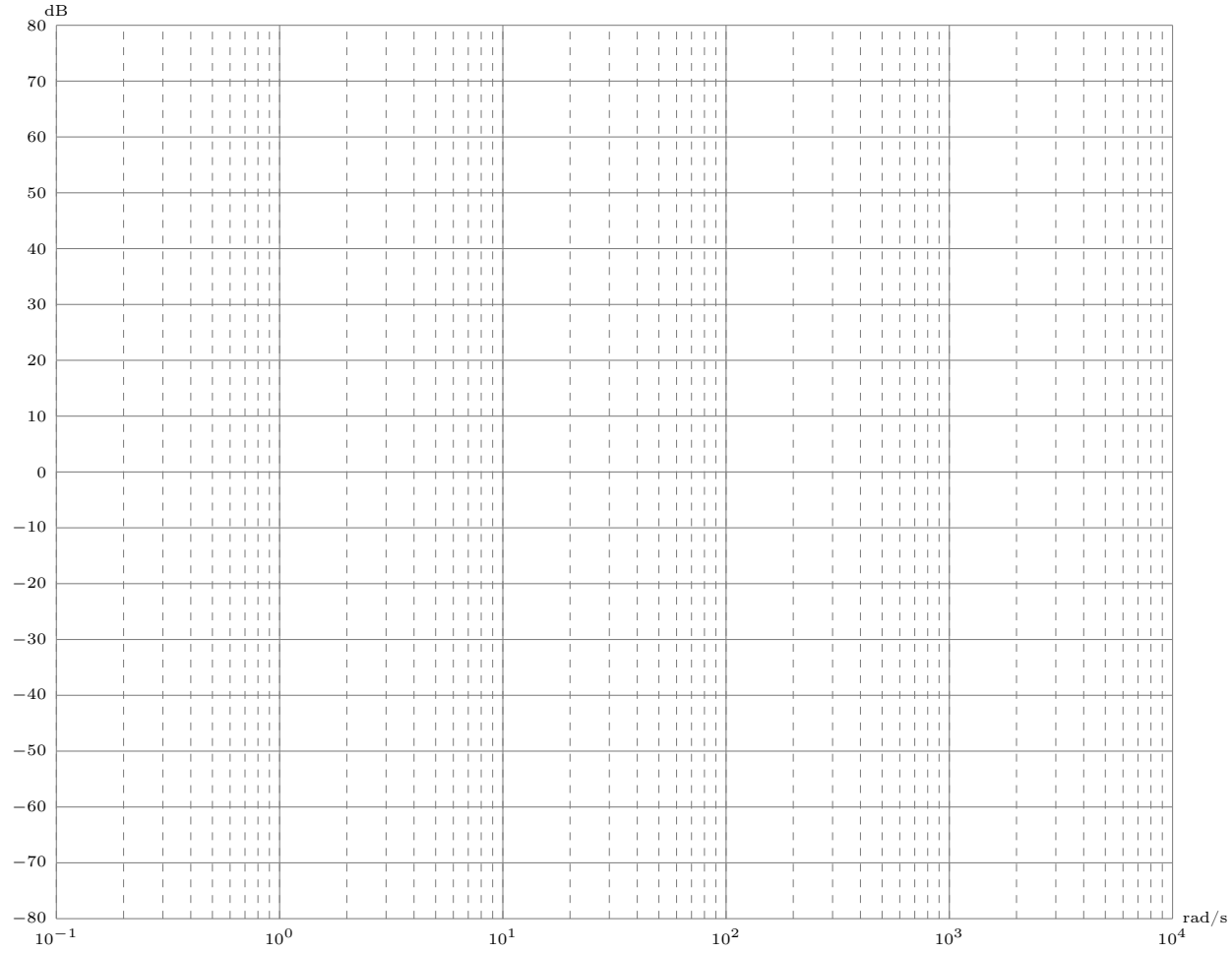

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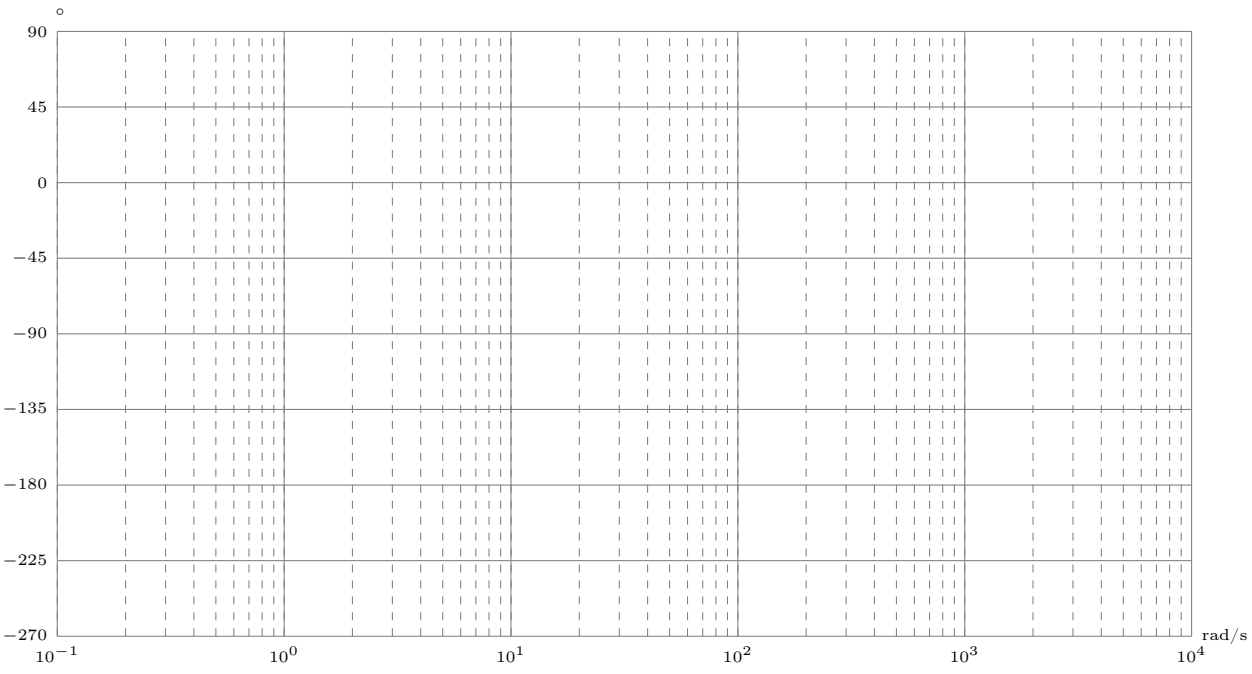

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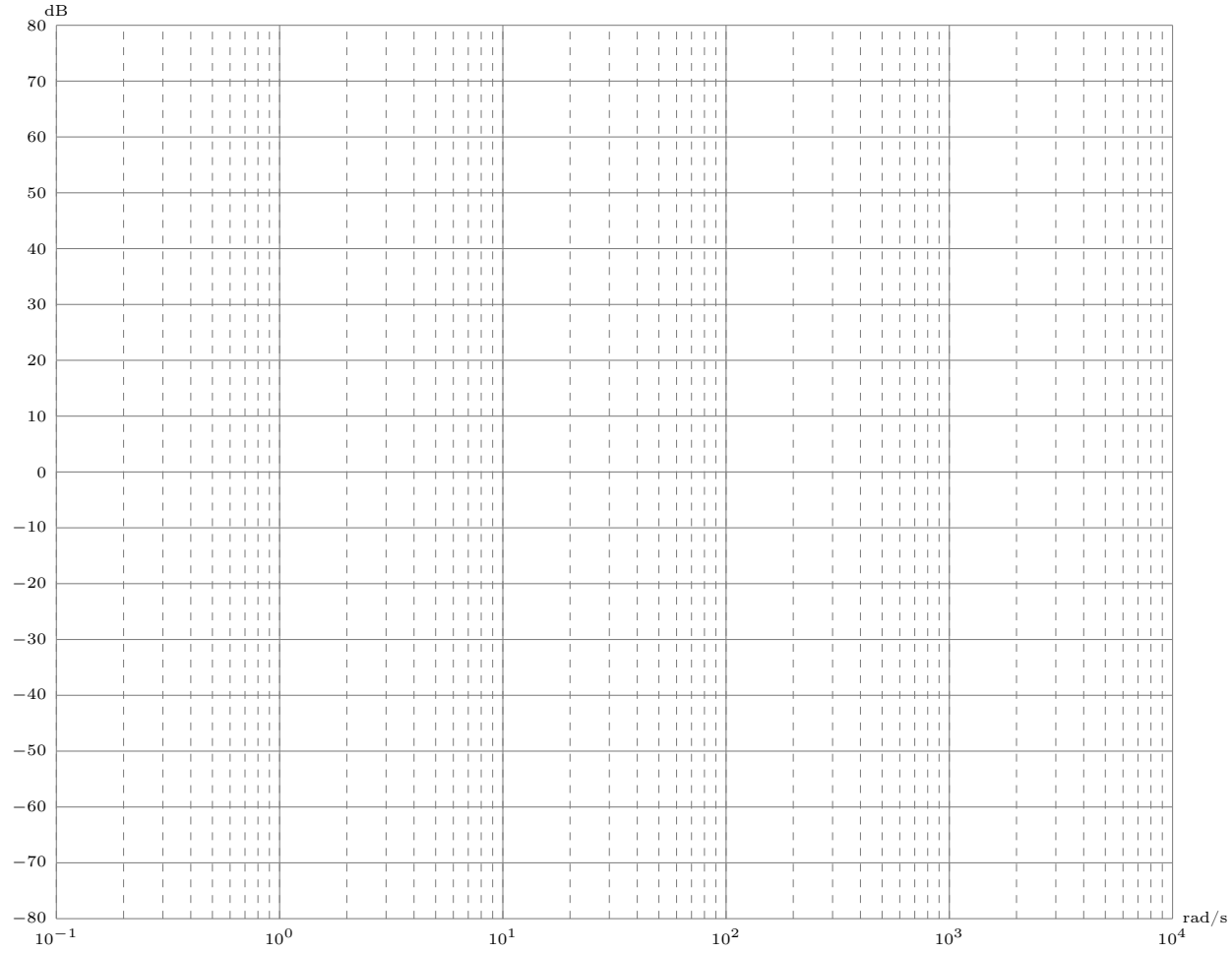

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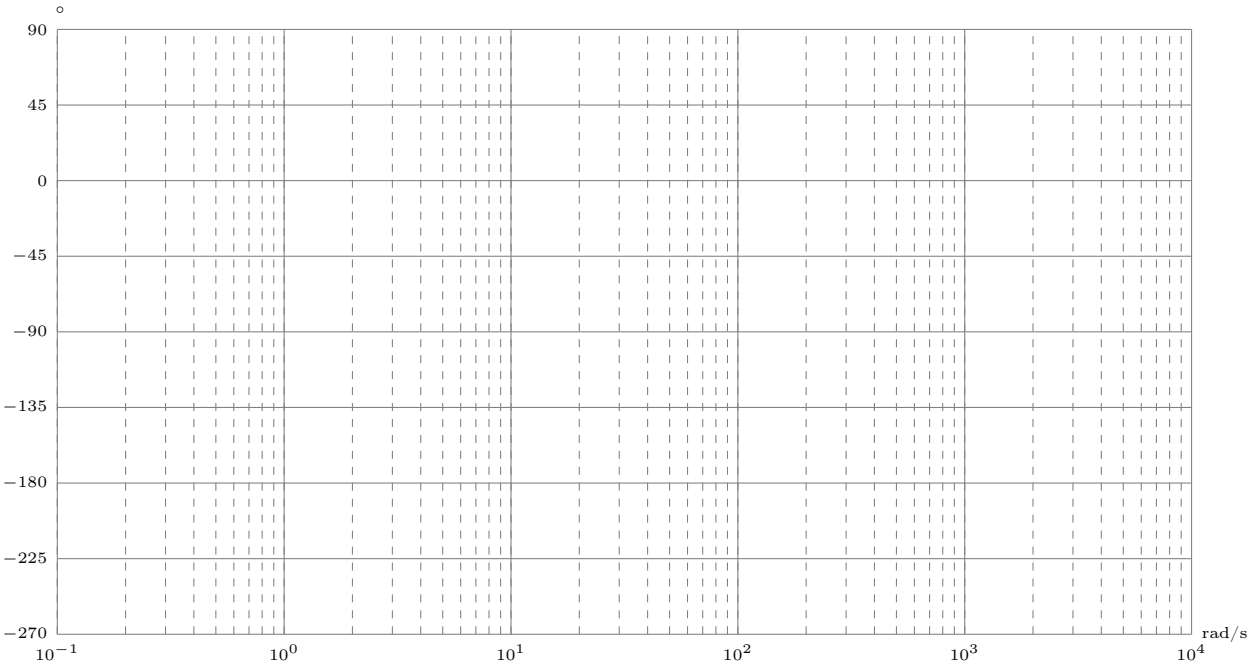

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